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United States Patent	12405086
Kind Code	B2
Date of Patent	September 02, 2025
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Adjustable reticles for sighting operations, and related methods, apparatus, and sight devices

Abstract

The present disclosure relates to adjustable reticles for sighting operations, and related methods, apparatus, and sight devices. In one or more embodiments, a sight device includes an illuminator. The illuminator includes a first illuminator section operable to illuminate a first section of a reticle, and a second illuminator section operable to illuminate a second section of the reticle radially outwardly of the first section. An outer dimension of the first section is a first ratio of an inner dimension of the second section, and the first ratio is less than 1:3.

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Appl. No.:	**18/416796**
Filed:	**January 18, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20250237480 A1	Jul. 24, 2025

Publication Classification

Int. Cl.: **F41G1/34** (20060101)

U.S. Cl.:

CPC **F41G1/345** (20130101);

Field of Classification Search

CPC: F42G (1/345)

USPC: 42/132

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Background/Summary

BACKGROUND
Field
(1) The present disclosure relates to adjustable reticles for sighting operations, and related methods, apparatus, and sight devices.
Description of the Related Art
(2) Reticles used in sighting operations (such as shooting operations) can be limited with respect to modularity such that it can be difficult for users to sight targets at a variety of distances and a variety of speeds. For example, it can be difficult for users to shoot targets at a variety of distances, a variety of shot split times, a variety of target transition speeds, and a variety of target acquisition speeds. Reticles can also be difficult for a user to align and/or level for accurate sighting. Moreover, reticles used in sighting operations can be distorted, blurry, and/or obstructive for a target, which hinders a user's accuracy and/or speed.
(3) Therefore, a need exists for improved reticles and sight devices.
SUMMARY

(4) The present disclosure relates to adjustable reticles for sighting operations, and related methods, apparatus, and sight devices.
(5) In one or more embodiments, a sight device includes an illuminator. The illuminator includes a first illuminator section operable to illuminate a first section of a reticle, and a second illuminator section operable to illuminate a second section of the reticle radially outwardly of the first section. An outer dimension of the first section is a first ratio of an inner dimension of the second section, and the first ratio is less than 1:3.
(6) In one or more embodiments, a sight device includes an illuminator. The illuminator includes a first illuminator section operable to illuminate a first section of a reticle, and a second illuminator section operable to illuminate a second section of the reticle radially outwardly of the first section at a spacing from the first section. The second section defines a gap radially outwardly of the spacing. A dimension of the gap is less than an outer dimension of the first section.
(7) In one or more embodiments, a non-transitory computer readable medium includes instructions that when executed cause a plurality of operations to be conducted. The plurality of operations include initiating an illumination of a first section of a reticle, and initiating an illumination of a second section of the reticle radially outwardly of the first section after the initiating of the illumination of the first section. The illumination of the second section is initiated while the first section is illuminated. The plurality of operations include ending the illumination of the second section, and initiating an illumination of a point section of the reticle radially outwardly of the second section after the ending of the illumination of the second section. The point section includes an outer point, and the illumination of the point section is initiated while the first section is illuminated.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of its scope, may admit to other equally effective embodiments.
(2) FIG. **1** is a schematic back perspective view of a sight device, according to one or more embodiments.
(3) FIG. **2** is a schematic front perspective view of the sight device shown in FIG. **1**, according to one or more embodiments.
(4) FIG. **3** is a schematic back perspective view of the sight device and a cover coupled to the sight housing **101**, according to one or more embodiments.
(5) FIG. **4** is a schematic front perspective view of the sight device and the cover shown in FIG. **3**, according to one or more embodiments.
(6) FIG. **5** is a schematic front view of the sight device shown in FIGS. **1-4**, according to one or more embodiments.
(7) FIG. **6** is a schematic cross-sectional view of the sight device and the cover along Section **6-6** shown in FIG. **3**, according to one or more embodiments.
(8) FIG. **7** is a schematic back perspective view of a sight device, according to one or more embodiments.
(9) FIG. **8** is a schematic front view of the sight device shown in FIG. **7**, according to one or more embodiments.
(10) FIG. **9** is a schematic diagram view of a reticle, according to one or

(11) more embodiments.
(12) FIG. **10** is a schematic side view of the illuminator shown in FIG. **2**, according to one or more embodiments.
(13) FIG. **11** is a schematic partial circuit diagram view of the illuminator and the PCB of the controller shown in FIG. **10**, according to one or more embodiments.
(14) FIGS. **12**A-**12**E is a schematic process flow view of a method of illuminating the reticle on the sight device, according to one or more embodiments.
(15) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

(16) The present disclosure relates to adjustable reticles for sighting operations, and related methods, apparatus, and sight devices. In one or more embodiments, a sight device is operable to view a reticle that is adjustable to increase and/or decrease a size of a central region (such as a central dot) of the reticle. The sight devices can be mounted to a variety of devices where aiming is used in operation of the devices. For example, the sight devices can be mounted to projectile launching devices, such as firearms (e.g., pistols, rifles, and/or shotguns), airguns, bows (e.g., crossbows), and/or airsoft guns. The present disclosure contemplates that the sight devices described herein can be mounted to other devices.
(17) The disclosure contemplates that terms such as “couples,” “coupling,” “couple,” and “coupled” may include but are not limited to bonding, embedding, welding, fusing, melting together, interference fitting, and/or fastening such as by using fasteners such as bolts, threaded connections, pins, and/or screws. The disclosure contemplates that terms such as “couples,” “coupling,” “couple,” and “coupled” may include but are not limited to integrally forming. The disclosure contemplates that terms such as “couples,” “coupling,” “couple,” and “coupled” may include but are not limited to direct coupling and/or indirect coupling, such as indirect coupling through components such as links, plates, blocks, and/or frames.
(18) FIG. **1** is a schematic back perspective view of a sight device **100**, according to one or more embodiments.
(19) FIG. **2** is a schematic front perspective view of the sight device **100** shown in FIG. **1**, according to one or more embodiments.
(20) FIGS. **1** and **2** are described together. The sight device **100** includes a sight housing **101** at least partially supporting a view panel **102** and an illuminator **103**. The view panel **102** can include, for example, a lens (such as a glass panel). The view panel **102** is curved in shape. In one or more embodiments, the view panel **102** includes a beam splitter. The beam splitter can transmit a first light (such as infrared radiation) and reflect a second light (such as visible light, e.g., visible red light or visible green light). For example, the view panel **102** can reflect visible light illuminated onto the view panel **102** using the illuminator **103**. The present disclosure contemplates that other view panels (such as displays) may be used in relation to the subject matter of the present application.
(21) The illuminator **103** is connected to a controller **110** that controls methods, such as the operations of the methods described herein. For example, the controller **110** controls the operation of the illuminator **103**. The controller **110** is disposed at least partially within the sight housing **101**. The controller **110** includes a central processing unit (CPU) **113** (e.g., a processor), a memory **111** containing instructions, and support circuits **112**. The controller **110** controls various items directly, or via other computers and/or controllers. In one or more embodiments, the controller **110** is communicatively coupled to dedicated controllers, and the controller **110** functions as a central controller.
(22) The controller **110** can include any form of a general-purpose computer processor that is used

for controlling sight devices, and sub-processors thereon or therein. The memory **111**, or non-transitory computer readable medium, can include one or more of a readily available memory such as random access memory (RAM), dynamic random access memory (DRAM), static RAM (SRAM), and synchronous dynamic RAM (SDRAM (e.g., DDR1, DDR2, DDR3, DDR3L, LPDDR3, DDR4, LPDDR4, and the like)), read only memory (ROM), floppy disk, hard disk, flash drive, or any other form of digital storage, local or remote. In one or more embodiments, the memory **111** includes a chip, for example a chip-on-board, such as a chip including silicon and/or germanium. The support circuits **112** of the controller **110** are coupled to the CPU **113** and/or the illuminator **103** for supporting the CPU **113** and/or the illuminator **103**. The support circuits **112** can include cache, power supplies, clock circuits, input/output circuitry and subsystems, and the like. Operational parameters (e.g., brightness settings) and operations are stored in the memory **111** as a software routine that is executed or invoked to turn the controller **110** into a specific purpose controller to control the operations of the sight devices described herein. The controller **110** is configured to conduct any of the operations described herein. The instructions stored on the memory **111**, when executed, cause one or more of the operations described herein to be conducted in relation to the sight device **100**.
(23) In one or more embodiments, the CPU **113**, the memory **111**, and the support circuits **112** are part of a printed circuit board (PCB). The illuminator **103** can be mounted to the PCB or spaced from the PCB while connected to the PCB.
(24) The various operations described herein can be conducted automatically using the controller **110**, or can be conducted automatically and/or manually with certain operations conducted by a user. As an example, a user can press and hold a first button **104** for a predetermined time (such as 2-3 seconds) to initiate a signal to the controller **110**, and in response to the signal the controller **110** turns on the illuminator **103**. As another example, a user can press the first button **104** to initiate a signal to the controller **110**, and in response to the signal the controller **110** increases a brightness setting for the illuminator **103**. As another example, a user can press a second button **105** to initiate a signal to the controller **110**, and in response to the signal the controller **110** decreases a brightness setting for the illuminator **103**. As a further example, a user can press and hold the second button **105** for a predetermined time (such as 2-3 seconds) to initiate a signal to the controller **110**, and in response to the signal the controller **110** turns off the illuminator **103**. As an additional example, a user can press and hold the first button **104** for a predetermined time (such as 2-3 seconds) while the illuminator **103** is on to initiate a signal to the controller **110**, and in response to the signal the controller **110** cycles the illuminator **103** through a plurality of operations (such as the operations shown in FIGS. **12**A-**12**E) until the first button **104** is released. In one or more embodiments, the first button **104** and the second button **105** are triangular in shape. In one or more embodiments, the first button **104** is oriented in a first direction to point toward the view panel **102** (e.g., in an upward direction), and the second button **105** is oriented in an opposite second direction to point away from the view panel **102** (e.g., in a downward direction).
(25) A first adjustment knob **106** can adjust a first setting (e.g., a vertical positon, such as an elevation) of the illuminator **103**, and a second adjustment knob **107** can adjust a second setting (e.g., a horizontal positon, such as a windage) of the illuminator **103**. A battery compartment **108** covers and/or supports a battery that supplies electrical power to the controller **110** and the illuminator **103**. The sight housing **101** includes a panel section **101***a* disposed about the view panel **102** and a flange section **101***b* extending relative to the panel section **101***a*. The flange section **101***b* includes a first outer side surface **115** and a second outer side surface **116** opposing the first outer side surface **115**. The flange section **101***b* includes a plurality of tapered outer surfaces **131**, **132** extending between the outer side surfaces **115**, **116** and a top surface **129** of the flange section **101***b*. A first fastener opening **117** is formed in the first outer side surface **115**, and a second fastener opening **118** is formed in the second outer side surface **116**. One or more fasteners **119**, **120** (two are shown) extend through the sight housing **101** to couple the sight housing **101** to a

firearm component (such as a slide, a frame, a receiver, and/or a rail system of a firearm) or a mount that couples to the firearm component. As an example, the fasteners **119**, **120** can thread into a slide, a receiver, or a frame of a firearm. As another example, the fasteners **119**, **120** can thread into a mount that couples to a firearm component. In one or more embodiments, the one or more fasteners **119**, **120** include screws.

(26) FIG. **3** is a schematic back perspective view of the sight device **100** and a cover **300** coupled to the sight housing **101**, according to one or more embodiments.

(27) FIG. **4** is a schematic front perspective view of the sight device **100** and the cover **300** shown in FIG. **3**, according to one or more embodiments.

(28) FIGS. **3** and **4** are described together. The cover **300** includes a cover housing **301** at least partially supporting a second view panel **302**. The cover housing **301** includes a panel section **301***a* disposed about the second view panel **302** and a flange section **301***b* extending relative to the panel section **301***a*. The flange section **301***b* of the cover housing **301** interfaces with (e.g., abuts against) the panel section **101***a* of the sight housing **101** when the cover housing **301** is supported on the sight housing **101**. The second view panel **302** is flat in shape and allows the visible light reflected by the view panel **102** to transmit therethrough and to a user's eye. The cover **300** facilitates preventing fluid (e.g., sweat and/or water, such as rain) and debris (such as dust, dirt, silt, and/or clothing lint) from obstructing the view panel **102** and/or the illuminator **103**. The cover **300** includes a first extension **304** (e.g., a first plate) and a second extension **305** (e.g., a second plate) extending relative to the panel section **301***a* of the cover housing **301**. The first extension **304** and the second extension **305** are spaced apart from each other by the flange section **101***b* of the sight housing **101**. At least part of the panel section **301***a* of the cover housing **301** extends between the first extension **304** and the second extension **305**. The first extension **304** extends along the first outer side surface **115** and the second extension **305** extends along the second outer side surface **116**. The cover **300** includes a first opening **308** formed in the first extension **304** and a second opening **309** formed in the second extension **305**. A first fastener **310** (e.g., a first screw) disposed in the first opening **308** extends through the first extension **304** and threads into the first fastener opening **117** of the sight housing **101**. A second fastener **311** (e.g., a second screw) disposed in the second opening **309** extends through the second extension **305** and threads into the second fastener opening **118** of the sight housing **101**. The first and second fasteners **310**, **311**. The first and second fasteners **310**, **311** are oriented parallel to the second view panel **302**.

(29) The cover **300** includes a plurality of tapered surfaces **331**, **332** extending between the extensions **304**, **305** and the panel section **301***a*. The tapered surfaces **331**, **332** interface with the tapered outer surfaces **131**, **132** of the sight housing **101** to facilitate guiding the cover **300** onto the sight housing **101** and retaining the cover **300** on the sight housing **101** during operation.

(30) The extensions **304**, **305** and/or the orientations of the first and second fasteners **310**, **311** facilitate reliable coupling of the cover **300** to the sight housing **101** throughout operation of the firearm to which the sight device **100** is mounted. For example, the extensions **304**, **305** and/or the orientations of the first and second fasteners **310**, **311** reduce or eliminate the loosening and dismounting of the cover **300** from the sight housing **101** under multiple recoil impulses over time. The extensions **304**, **305** and/or the orientations of the first and second fasteners **310**, **311** facilitate a reduced footprint for the sight device **100**, which facilitates enhanced modularity and reduced interference with other firearm components (such as mechanical sights). The extensions **304**, **305** and/or the orientations of the first and second fasteners **310**, **311** facilitate ease of installation and removal of the cover **300** to and from the sight housing **100** while the sight device **100** is mounted to the firearm, such as when mechanical sight(s) of the firearm are adjacent to or near the sight device **100**.

(31) FIG. **5** is a schematic front view of the sight device **100** shown in FIGS. **1-4**, according to one or more embodiments.

(32) The panel section **101***a* of the sight housing **101** includes a plurality of outer surfaces **125**

(eleven are shown in FIG. **5**). A different number (lower or higher) of the outer surfaces **125** can be used. One or more additional outer surfaces can be used that are oriented differently than the outer surfaces **125**. The outer surfaces **125** are disposed about the view panel **102**. In one or more embodiments, the outer surfaces **125** are planar. The outer surfaces **125** are oriented to intersect each other and form a plurality of outer edges **126** between adjacent outer surfaces **125**. The outer surfaces **125** are oriented such that an angle A**1** is defined between adjacent outer surfaces **125**. At least two (such as two or all) of the outer surfaces **125** are oriented to define the angle A**1** between adjacent outer surfaces **125**. The angle A**1** is less than 35 degrees, such as 30 degrees or less. In one or more embodiments, the angle A**1** is less than 30 degrees. In one or more embodiments, the angle A**1** is within a range of 20 degrees to 30 degrees. In one or more embodiments, the angle A**1** is within a range of 24 degrees to 28 degrees, such as about 26 degrees. The outer surfaces **125** facilitate enhancing durability (e.g., impact resistance and/or reduced wear) and operational lifespans of the sight device **100** while enhancing the aesthetic appeal of the sight device **100**.
(33) One or more alignment tabs **127**, **128** (two are shown) can be used to align the sight housing **101** relative to the mount or firearm component prior coupling the sight housing **101** to the mount or firearm component using the one or more fasteners **119**, **120**.
(34) FIG. **6** is a schematic cross-sectional view of the sight device **100** and the cover **300** along Section **6-6** shown in FIG. **3**, according to one or more embodiments.
(35) The panel section **301***a* of the cover housing **301** includes a plurality of outer surfaces **325** (nine are shown in FIG. **6**). A different number (lower or higher) of the outer surfaces **325** can be used. One or more additional outer surfaces can be used that are oriented differently than the outer surfaces **325**. The outer surfaces **325** are disposed about the second view panel **302**. The outer surfaces **325** align with the outer surfaces **125**. The outer surfaces **325** are similar to the outer surfaces **125** and include one or more aspects, features, properties, and/or operations thereof. For example, the outer surfaces **325** are oriented such that the angle A**1** is defined between adjacent outer surfaces **325**. The outer surfaces **325** facilitate enhancing durability (e.g., impact resistance and/or reduced wear) and operational lifespans of the cover **300** while enhancing the aesthetic appeal of the cover **300**.
(36) FIG. **7** is a schematic back perspective view of a sight device **700**, according to one or more embodiments.
(37) FIG. **8** is a schematic front view of the sight device **700** shown in FIG. **7**, according to one or more embodiments.
(38) FIGS. **7** and **8** are described together. The sight device **700** is similar to the sight device **100** shown in FIGS. **1-6** and includes one more aspects, features, components, operations, and/or properties thereof. For example, the sight device **700** can include the illuminator **103** and/or the controller **110** among other features and components.
(39) The sight device **700** includes a sight housing **701** that includes a panel section **701***a* and a flange section **701***b*.
(40) The panel section **701***a* of the sight housing **701** includes a plurality of outer surfaces **725** (seven are shown in FIG. **7**). A different number (lower or higher) of the outer surfaces **725** can be used. One or more additional outer surfaces can be used that are oriented differently than the outer surfaces **725**. The outer surfaces **725** are disposed about the view panel **102**. In one or more embodiments, the outer surfaces **725** are planar. The outer surfaces **725** are oriented such that an angle A**2** is defined between adjacent outer surfaces **725**. At least two (such as two or all) of the outer surfaces **725** are oriented to define the angle A**2** between adjacent outer surfaces **725**. The angle A**2** is less than 35 degrees, such as 30 degrees or less. In one or more embodiments, the angle A**2** is less than 20 degrees. In one or more embodiments, the angle A**2** is within a range of 10 degrees to 20 degrees. In one or more embodiments, the angle A**2** is within a range of 13 degrees to 17 degrees, such as about 15 degrees.
(41) FIG. **9** is a schematic diagram view of a reticle **900**, according to one or more embodiments.

Sight device(s) can be operable to view the reticle **900**. The reticle **900** can be illuminated as visible light using the illuminator **103** onto the view panel **102** of the sight device **100** or the sight device **700**. Although the sight devices **100**, **700** are described as reflex sights, the reticle **900** can be illuminated and/or viewable on a variety of other sight devices. For example, the reticle **900** can be illuminated and/or viewable as part of a wire reticle, an etched reticle, and/or a fiber reticle. As another example, the reticle **900** can be at least partially illuminated through a view panel to a user's eye(s) in addition to or in place of reflecting the reticle **900** at least partially off of the view panel **102**. As a further example, the reticle **900** can be illuminated and/or viewable as part of a holographic sight and/or a magnifying scope. As another example, the reticle **900** can be illuminated and/or viewable on a display, such as a display screen (for example an organic light-emitted diode (OLED) display screen). Other sight devices are contemplated. The present disclosure contemplates that the reticle **900** can be viewable in a manner that is not illuminated. For example, the reticle **900** can be a wire reticle and/or an etched reticle.
(42) The reticle **900** includes a first section **910**, a second section **930** disposed radially outwardly of the first section **910**, and a third section **950** disposed radially outwardly of the second section **930**. The sections **910**, **930**, **950**, **970** of the reticle **900** are shown with surface shading in FIG. **9** for visual clarity purposes. The sections **910**, **930**, **950**, **970** of the reticle **900** can be filled with solid color(s). For example, the sections **910**, **930**, **950**, **970** of the reticle **900** can be filled with visible red light and/or visible green light.
(43) An outer dimension OD**1** of the first section **910** is a first ratio of an inner dimension ID**1** of the second section **930**, and the first ratio is less than 1:3 (e.g., less than about 0.33333). In one or more embodiments, the first ratio is 0.31 or less, such as within a range of 0.29 to 0.31. A spacing **911** is between the outer dimension OD**1** of the first section **910** and the inner dimension ID**1** of the second section **930**. The spacing **911** is a second ratio of the outer dimension OD**1** of the first section, and the second ratio is greater than 1.0. The second ratio is less than 1.5, such as less than 1.3. In one or more embodiments, the second ratio is within a range of 1.10 to 1.2. In one or more embodiments, the second ratio is within a range of 1.15 to 1.17, such as about 1.16071. The spacing **911** surrounds the first section **910**. The second section **930** partially surrounds the first section **910**.
(44) The spacing **911** is a third ratio of an inner dimension ID**2** of the third section **950**, and the third ratio is 0.05 or less. In one or more embodiments, the third ratio is within a range of 0.04 to 0.05. In one or more embodiments, the third ratio is within a range of 0.045 to 0.047, such as about 0.046. An outer dimension OD**2** of the second section **930** is a fourth ratio of the outer dimension OD**1** of the first section **910**, and the fourth ratio is greater than 5.0. In one or more embodiments, the fourth ratio is within a range of 5.05 to 5.15, such as about 5.1.
(45) The outer dimension OD**1** of the first section **910** is a fifth ratio of an inner dimension ID**2** of the third section **950**, and the fifth ratio is 0.05 or less. In one or more embodiments, the fifth ratio is within a range of 0.035 to 0.05. In one or more embodiments, the fifth ratio is within a range of 0.035 to 0.045, such as about 0.04. The present disclosure contemplates that the use of "about" herein can encompass a difference of 5% or less. The inner dimension ID**1** of the second section **930** is a sixth ratio of the inner dimension ID**2** of the third section **950**, and the sixth ratio less than 0.2. In one or more embodiments, the sixth ratio is within a range of 0.1 to 0.15, such as about 0.13.
(46) The second section **930** defines a gap **931** radially outwardly of the spacing **911**. A dimension D**1** of the gap **931** is less than the outer dimension OD**1** of the first section **910**. The dimension D**1** of the gap **931** is a seventh ratio of the outer dimension OD**1** of the first section OD**1**, and the seventh ratio is less than 1.5. In one or more embodiments, the seventh ratio is greater than 0.4 and less than 1.0. In one or more embodiments, the seventh ratio is less than 0.6. In one or more embodiments, the seventh ratio is within a range of 0.5 to 0.55, such as within a range of 0.53 to 0.54. The dimension D**1** of the gap **931** is an eighth ratio of the spacing **911**, and the eighth ratio is

less than 1.3. In one or more embodiments, the eighth ratio is less than 1.0, such as less than 0.6 or less than 0.5. In one or more embodiments, the eighth ratio is within a range of 0.4 to 0.5, such as within a range of about 0.45 to 0.47. The dimension D**1** of the gap **931** is a ninth ratio of the inner dimension ID**1** of the second section **930**, and the ninth ratio is less than 0.5. In one or more embodiments, the ninth ratio is less than 0.3, such as less than 0.2. In one or more embodiments, the ninth ratio is within a range of 0.1 to 0.2, such as within a range of about 0.15 to 0.17.

(47) The dimension D**1** of the gap **931** is a tenth ratio of the inner dimension ID**2** of the third section **950**, and the tenth ratio is less than 0.06. In one or more embodiments, the tenth ratio is less than 0.05, such as less than 0.03. In one or more embodiments, the tenth ratio is within a range of 0.01 to 0.03, such as within a range of about 0.015 to 0.025. In one or more embodiments, the tenth ratio is within a range of 0.020 to 0.022, such as about 0.021.

(48) The reticle **900** includes a fourth section **970** at least partially aligned with the third section **950** at a radial location. The third section **950** at least partially defines a gap **954**, and the fourth section **970** is disposed at least partially in the gap **954**. The fourth section **970** includes an outer point **971** disposed radially between the third section **950** and the second section **930**. In one or more embodiments, the outer point **971** is an apex of the fourth section **970**. The outer point **971** is disposed at a radial distance R**1**. The radial distance R**1** is an eleventh ratio of the inner dimension ID**2** of the third section **950**. The eleventh ratio is less than 0.5 and equal to or greater than 0.4. The radial distance R**1** is a twelfth ratio of the outer dimension OD**2** of the second section **930**, and the twelfth ratio is at least 2.0. The gap **931** of the second section **930** is aligned radially between the first section **910** and the outer point **971** of the fourth section **970**. The gap **954** of the third section **950** has a dimension D**2**. The fourth section **970** has a first dimension D**3** and a second dimension D**4** oriented perpendicularly to the first dimension D**3**.

(49) The spacing **911** is a thirteenth ratio of the inner dimension ID**1** of the second section **930**. The thirteenth ratio is less than 0.5, such as less than 0.4. In one or more embodiments, the thirteenth ratio is greater than 1:3 (e.g., greater than about 0.33333). In one or more embodiments, the thirteenth ratio is within a range of 0.30 to 0.40, such as about 0.3495.

(50) The third section **970** includes an outer dimension OD**3**. In one or more embodiments, the third section **950** includes an arcuate portion **951**, a first side portion **952** extending linearly relative to a first side of the arcuate portion **951**, and a second side portion **953** extending linearly relative to a second side of the arcuate portion **951**. The first and second side portions **952**, **953** respectively have a thickness T**1**. In one or more embodiments, the dimension D**2** is between two end points of the arcuate portion **951**, and the two end points are in the middle of a thickness T**2** of the arcuate portion **951**. The reticle **900** has an overall height H**1** and an overall width W**1**.

(51) In one or more embodiments, the outer dimension OD**1** is an outer diameter of the first section **910**, the inner dimension ID**1** is an inner diameter of the second section **930**, the outer dimension OD**2** is an outer diameter of the second section **930**, the inner dimension ID**2** is an inner diameter of the third section **950**, and/or the outer dimension OD**3** is an outer diameter of the third section **950**. In one or more embodiments, the dimension D**1** is a width of the gap **931**, the dimension D**2** is a width of the gap **954**, the first dimension D**3** is a width of the fourth section **970**, and/or the second dimension D**4** is a height of the fourth section **970**.

(52) In one or more embodiments, the first section **910** includes a dot, the second section **930** includes a first arc, the third section **950** includes a second arc, and/or the fourth section **970** includes a triangle. The present disclosure contemplates that other shapes may be used. For example, the fourth section **970** can include a chevron, an arc, or a star. Other shapes are contemplated.

(53) The outer dimension OD**1** of the first section **910** is within a range of 25 microns to 30 microns, such as about 28 microns. In one or more embodiments, the outer dimension OD**1** of the first section **910** is about 2.5 minutes-of-angle (MOA). The inner dimension ID**1** of the second section **930** is 100 microns or less. The inner dimension ID**1** of the second section **930** is within a

range of 85 microns to 100 microns, such as about 93 microns. In one or more embodiments, the inner dimension ID**1** of the second section **930** is about 8 MOA. The outer dimension OD**2** of the second section **930** is within a range of 130 microns to 155 microns, such as about 143 microns. In one or more embodiments, the outer dimension OD**2** of the second section **930** is about 12.5 MOA. The inner dimension ID**2** of the third section **950** is within a range of 700 microns to 720 microns, such as about 707 microns. In one or more embodiments, the inner dimension ID**2** of the third section **950** is about 62 MOA. The outer dimension OD**3** of the third section **950** is within a range of 750 microns to 790 microns, such as about 767 microns. In one or more embodiments, the outer dimension OD**3** of the third section **950** is about 68 MOA. The spacing **911** is 30 microns or less. The spacing **911** is within a range of 25 microns to 30 microns, such as about 32.5 microns. In one or more embodiments, the spacing **911** is about 2.75 MOA. The dimension D**1** of the gap **931** is 20 microns or less. The dimension D**1** of the gap **931** is within a range of 10 microns to 20 microns, such as about 0.015 microns. In one or more embodiments, the dimension D**1** of the gap **931** is about 1.3 MOA. In one or more embodiments, the spacing **911** is omitted. In such an embodiment, the inner dimension ID**1** of the second section **930** can be equal to the outer dimension OD**1** of the first section **910**. In one or more embodiments, the gap **931** is omitted. In such an embodiment, the second section **930** can include a complete ring (e.g., a complete circular ring) that surrounds the first section **910**. In one or more embodiments, the gap **954** is omitted on one or both sides of the fourth section **970**. In such an embodiment, the arcuate portion **951** of the third section **950** can connect to the fourth section **970** on one or both sides of the fourth section **970**. In one or more embodiments, the fourth section **970** is omitted and/or the arcuate portion **951** of the third section **950** includes a complete ring (e.g., a complete circular ring) that surrounds the second section **930**.
(54) The dimension D**2** of the gap **954** is within a range of 240 microns to 265 microns, such as about 252 microns. In one or more embodiments, the dimension D**2** of the gap **954** is about 21 MOA. The first dimension D**3** of the fourth section **970** is within a range of 100 microns to 120 microns, such as about 107 microns. In one or more embodiments, the first dimension D**3** is about 9 MOA. The second dimension D**4** of the fourth section **970** is within a range of 110 microns to 130 microns, such as about 117 microns. In one or more embodiments, the second dimension D**4** is about 10 MOA. The radial distance R**1** is within a range of 290 microns to 320 microns, such as about 305 microns. In one or more embodiments, the radial distance R**1** is about 26 MOA. The thickness T**1** is within a range of 25 microns to 35 microns, such as about 30 microns. In one or more embodiments, the thickness T**1** is about 2.5 MOA. The overall width W**1** is within a range of 940 microns to 965 microns, such as about 953 microns. In one or more embodiments, the overall width W**1** is about 84 MOA. The overall height H**1** is within a range of 790 microns to 820 microns, such as about 805 microns. In one or more embodiments, the overall height H**1** is about 71 MOA. A length L**1** of the respective first side portion **952** and second side portion **953** is within a range of 150 microns to 220 microns, such as 180 microns to 190 microns, for example about 186 microns. In one or more embodiments, the length L**1** is within a range of 6 MOA to 10 MOA, such as about 8 MOA.
(55) A first parameter of the first section **910** is controllable independently of a second parameter of the second section **930**. A third parameter of the third section **950** is controllable independently of the first and second parameters. A fourth parameter of the fourth section **970** is controllable independently of the first, second, and third parameters. The parameters can respectively include, for example, an on/off setting, a brightness (e.g., a light intensity), and/or a color. Other parameters are contemplated.
(56) The present disclosure contemplates that the ordered numerals for items described herein are for clarity purposes and the items can be labeled with other numerals depending on the number of items present. For example, the fifth ratio described above can be referred to as a “first ratio” or a “second ratio,” and/o the sixth ratio described above can be referred to as a “second ratio” or a “third ratio” if used in addition to the fifth ratio. As another example, the seventh ratio, the eighth

ratio, the ninth ratio, the tenth ratio, the eleventh ratio, the twelfth ratio, and/or the thirteenth ratio described above can be respectively referred to as "a ratio" or a "first ratio."

(57) The reticle **900** involves an adjustable multi-reticle. For example, the first section **910** and the second section **930** involve an adjustable dot such that the dot of the first section **910** appears larger when the second section **930** is illuminated in addition to the first section **910**. For example, the spacing **911** and the gap **931** appear omitted in the sight picture to the user. The spacing **911**, the outer dimension OD**1** of the first section **910**, and the inner dimension ID**1** of the second section **930**—and the associated ratios for these aspects—facilitate the larger dot in a manner that is clear and crisp with reduced or eliminated distortion and blurriness of the larger dot, while facilitating reliable operation and ease of manufacturing. For example, interference between the illuminated second section **930** and the illuminated first section **910** is reduced or eliminated while substantially maintaining the appearance of the shape of a dot. As another example, the appearance of the shape of the dot is clear and crisp during fast target transitions. Additionally, the gap **931**, the dimension D**1**—and the associated ratios for these aspects—facilitate the larger dot in the manner that is clear and crisp while facilitating ease of manufacturing and reliable operation of the illuminator **103**. For example, the appearance of the shape of the dot is substantially maintained while reliably providing power to the sections of the illuminator **103** to reliably and independently turn the sections **910**, **930**, **950**, **970** on and off.

(58) The adjustable reticle **900** facilitates modularity in target acquisition and shooting. For example, the smaller dot (e.g., the first section **910** on with the second section **930** off) can be used for more precise shots at longer distances, and/or shots at smaller targets, whereas the larger dot (e.g., both the first section **910** and the second section **930** on) can be used for faster shots, shots at shorter distances, and/or shots at larger targets. As another example, the third section **950** (including the arcuate portion **951**, the first side portion **952**, and/or the second side portion **953**) can be used for quick target acquisition, leveling of the reticle **900**, and/or alignment of the user's eye(s) with a line of sight of the view panel **102**. The first side portion **952** and/or the second side portion **953** can be used for speed shooting (e.g., in shooting sports such as the Steel Challenge) and anticipating trigger sear engagement. For example, as a user moves the reticle **900** toward a target the user can complete a trigger pull as an outer edge **956** of the second side portion **953** reaches an outer edge of the target, and the trigger sear can be engaged after the shot and as the reticle **900** moves across the target and toward a second target. As a further example, the first section **910** and/or the second section **930** can be used to sight farther targets and the fourth section **970** can be used to sight closer targets. The first section **910** and/or the second section **930** can be zeroed for a distance (e.g., about 75 meters) and the outer point **971** can be used for targets at a closer distance (e.g., in a range of 0 meters to about 10 meters). Other distances are contemplated. The present disclosure contemplates that the larger dot can be illuminated in another manner. For example, at least one illuminator section of the illuminator **103** (such as the first illuminator section **1001** described below) can be moved to enlarge the first section **910** to have the outer dimension OD**2** shown for the second section **930**. For example, the illuminator **103** can be moved closer to the view panel **102** to increase the size of the first section **910**, and the illuminator **103** can be moved farther from the view panel **102** to decrease the size of the first section **910**. The illuminator **103** can be moved closed and farther from an aperture to increase or decrease the size of the first section **910**. The aperture can be between the illuminator **103** and the view panel **102**. The illuminator **130** can be moved, for example, using an adjustment knob that can be similar to the first adjustment knob **106** and/or the second adjustment knob **107**.

(59) FIG. **10** is a schematic side view of the illuminator **103** shown in FIG. **2**, according to one or more embodiments.

(60) The illuminator **103** is part of the controller **110**. For example, the illuminator **103** can be disposed on a side of a PCB **1110** of the controller **110**. The illuminator **103** includes a first illuminator section **1001** (e.g., a first light emitting diode (LED)) operable to illuminate the first

section **910** of a reticle **900**, a second illuminator section **1002** (e.g., a second LED) operable to independently illuminate the second section **930**, a third illuminator section **1003** (e.g., a third LED) operable to independently illuminate the third section **950** of the reticle **900**, and a fourth illuminator section **1004** (e.g., a fourth LED) operable to independently illuminate the fourth section **970** of the reticle **900**. In one or more embodiments, the same brightness setting is applied to the respective illuminator sections **1001**, **1002**, **1003**, and/or **1004** that are turned on. Other implementations are contemplated. The present disclosure contemplates that one LED or a plurality of LEDs can respectively be used for each of the illuminator sections **1001-1004**. The first illuminator section **1001** is similar in shape to the first section **910** of the reticle **900** described above. The second illuminator section **1002** is similar in shape to the second section **930** of the reticle **900** described above. The third illuminator section **1003** is similar in shape to the third section **950** of the reticle **900** described above. The fourth illuminator section **1004** is similar in shape to the fourth section **970** of the reticle **900** described above.

(61) The respective dimensions of the illuminator sections **1001-1004** of the illuminator **103** are equal to the respective dimensions described above for the sections **910**, **920**, **950**, **970** divided by a factor. As such, the dimensions of the illuminator sections **1001-1004** equal to the dimensions of the sections **910**, **920**, **950**, **970** scaled down by the factor. The respective ratios defining the dimensions of the illuminator sections **1001-1004** of the illuminator **103** are equal to the respective ratios described above for defining the dimensions of the sections **910**, **920**, **950**, **970** of the reticle **900**. For example, a spacing **1007** between the first illuminator section **1001** and the second illuminator section **1002** is the second ratio of the outer dimension of the first illuminator section **1001**, the third ratio of the inner dimension of the third illuminator section **1003**, and the thirteenth ratio of the second illuminator section **1002**. As another example, a gap **1006** defined by the second illuminator section **1002** is the seventh ratio of the outer dimension relative to the first illuminator section **1001**, the eighth ratio of the spacing **1007**, the ninth ratio of the inner dimension of the second illuminator section **1002**, and the tenth ratio of the inner dimension of the third illuminator section **1003**. As a further example, the first section **1001** includes the first ratio and the fifth ratio described above, the second section **1002** includes the fourth ratio and the sixth ratio described above, and the radial distance of the fourth section **1004** includes the eleventh ratio and the twelfth ratio described above.

(62) A first wire **1021** connects the first illuminator section **1001** to a first connector **1031** for power supply and power return to and from the first illuminator section **1001**. A second wire **1022** connects the second illuminator section **1002** to a second connector **1032** for power supply and power return to and from the second illuminator section **1002**. A third wire **1023** connects the third illuminator section **1003** to a third connector **1033** for power supply and power return to and from the third illuminator section **1003**. A fourth wire **1024** connects the fourth illuminator section **1004** to a fourth connector **1034** for power supply and power return to and from the fourth illuminator section **1004**. The first and second wires **1021**, **1022** extend through a gap **1008** defined by the third illuminator section **1003**. The first wire **1021** extends through the gap **1006** defined by the second illuminator section **1002**.

(63) FIG. **11** is a schematic partial circuit diagram view of the illuminator **103** and the PCB **1110** of the controller **110** shown in FIG. **10**, according to one or more embodiments. The processor **113** of the controller **110** controls a plurality of switches **1101**, **1102**, **1103**, **1104** (e.g., relays) to independently supply power respectively to the four illuminator sections **1001-1004** (e.g., the four LEDs). The switches **1101-1104** are respectively connected to the wires **1021-1024** through the connectors **1031-1034**. A power source **1111** is connected to the switches **1101-1104**. The power source **1111** can include, for example, a battery and/or a solar cell.

(64) FIGS. **12**A-**12**E is a schematic process flow view of a method of illuminating the reticle **900** on the sight device **100** or **700**, according to one or more embodiments. FIGS. **12**A-**12**E show the sequence of the sections **910**, **930**, **950**, **970** illuminated for the reticle **900**.

(65) At FIG. **12**A, a first operation includes initiating an illumination of the first section **910** of the reticle **900**. The first operation includes powering the first illuminator section **1001** to turn on the first section **910**.
(66) At FIG. **12**B, a second operation includes initiating an illumination of the second section **930** of the reticle **900** radially outwardly of the first section **910** after the initiating of the illumination of the first section **910** (FIG. **12**A). The illumination of the second section **930** initiated while the first section **910** is illuminated. The second operation includes powering the second illuminator section **1002** to turn on the second section **930**.
(67) At FIG. **12**C, a third operation includes ending the illumination of the second section **930**. The third operation includes ending power to the second illuminator section **1002** to turn off the second section **930**. Also at FIG. **12**C, a fourth operation includes initiating an illumination of the fourth section **970** (e.g., a point section) of the reticle **900** radially outwardly of the second section **930** after the ending of the illumination of the second section **930**. The illumination of the fourth section **970** is initiated while the first section **910** is illuminated. The fourth operation includes powering the fourth illuminator section **1004** to turn on the fourth section **970**. The third operation and fourth operation can be conducted simultaneously or sequentially with respect to each other.
(68) At FIG. **12**D, a fifth operation includes initiating an illumination of the third section **950** of the reticle **900** radially outwardly of the second section **930** after the initiating of the illumination of the fourth section **970** (FIG. **12**C). The illumination of the third section **970** is initiated while the first section **910** and the fourth section **970** are illuminated. The fifth operation includes powering the third illuminator section **1003** to turn on the third section **950**.
(69) At FIG. **12**E, a sixth operation includes re-initiating the illumination of the second section **930** of the reticle **900** after the initiating of the illumination of the third section **950** (FIG. **12**D). The illumination of the second section **930** is re-initiated while the first section **910**, the fourth section **970**, and the third section **950** are illuminated. The sixth operation includes powering the second illuminator section **1002** to turn on the second section **930**.
(70) The method of FIGS. **12**A-**12**E facilitate modularity in quickly cycling between a variety of reticle settings for a variety of shooting conditions (such as shooting speed, target size, and/or target distance) for competition shooting and other shooting applications. For example, the method can quickly cycle between reticle settings for shooting conditions that can arise in competition shooting applications, military shooting applications, law enforcement shooting applications, and/or recreational shooting applications.
(71) The subject matter of the present application can be expressed in one or more of the following Examples.
(72) Example 1 includes a reticle that includes: a first section; a second section disposed radially outwardly of the first section; a third section disposed radially outwardly of the second section; and a fourth section at least partially aligned with the third section at a radial location.
(73) Example 2 includes the reticle of Example 1, wherein the third section at least partially defines a gap, and the fourth section is disposed at least partially in the gap.
(74) Example 3 includes the reticle of Example 1, wherein the fourth section includes an outer point disposed radially between the third section and the second section.
(75) Example 4 includes the reticle of Example 3, wherein the outer point is an apex of the fourth section.
(76) Example 5 includes the reticle of Example 3, wherein the outer point is disposed at a radial distance, the radial distance is a ratio of an inner dimension of the third section, and the ratio is less than 0.5 and equal to or greater than 0.4.
(77) Example 6 includes the reticle of Example 5, wherein the inner dimension of the third section is an inner diameter.
(78) Example 7 includes the reticle of Example 3, wherein the outer point is disposed at a radial distance, the radial distance is a ratio of an outer dimension of the second section, and the ratio is at

least 2.0.
(79) Example 8 includes the reticle of Example 7, wherein the outer dimension of the second section is an outer diameter.
(80) Example 9 includes the reticle of Example 1, wherein the first section includes a dot, the second section includes a first arc, the third section includes a second arc, and the fourth section includes a triangle.
(81) Example 10 includes a sight device operable to view the reticle of Example 1.
(82) Benefits of the present disclosure include enhanced shooting accuracy; enhanced speed; adjustable reticles for sight modularity; sighting targets at a variety of distances and a variety of speeds (e.g., using a variety of shot split times, a variety of target transition speeds, and a variety of target acquisition speeds); and quick and easy aligning and/or leveling of reticles. Benefits of the present disclosure also include enhanced reticle clarity and sight clarity (such as reduced or eliminated distortion and blur for, e.g., an adjustable dot); and reduced or eliminated obstruction of targets. Such benefits can be further enhanced depending on user conditions, such as eye astigmatism.
(83) Benefits of the present disclosure also include reduced or eliminated obstruction of view panels and/or illuminators using a cover while facilitating reliable coupling of the cover to the sight device, reduced sight footprints, and enhanced sight modularity.
(84) It is contemplated that one or more aspects disclosed herein may be combined. As an example, one or more aspects, features, components, operations and/or properties of the sight device **100**; the illuminator **103**; the controller **110**; the outer surfaces **125**; the cover **300**; the outer surfaces **325**; the sight device **700**; the outer surfaces **725**; the reticle **900**; the PCB **1110**; and/or the method shown in FIGS. **12**A-**12**E may be combined. Moreover, it is contemplated that one or more aspects disclosed herein may include some or all of the aforementioned benefits.
(85) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A sight device, comprising: an illuminator comprising: a first illuminator section operable to illuminate a first section of a reticle, and a second illuminator section operable to illuminate a second section of the reticle radially outwardly of the first section, an outer dimension of the first section is a first ratio of an inner dimension of the second section, and the first ratio is less than 1:3.
2. The sight device of claim 1, wherein a first parameter of the first section is controllable independently of a second parameter of the second section.
3. The sight device of claim 1, wherein the first ratio is within a range of 0.29 to 0.31.
4. The sight device of claim 1, wherein a spacing is between the outer dimension of the first section and the inner dimension of the second section, the spacing is a second ratio of the outer dimension of the first section, and the second ratio is greater than 1.0.
5. The sight device of claim 4, wherein the second ratio is within a range of 1.10 to 1.2.
6. The sight device of claim 4, wherein the spacing surrounds the first section, and the second section partially surrounds the first section.
7. The sight device of claim 4, wherein the illuminator further comprises: a third illuminator section operable to illuminate a third section of the reticle radially outwardly of the second section, wherein the spacing is a third ratio of an inner dimension of the third section, and the third ratio is 0.05 or less.
8. The sight device of claim 1, wherein the illuminator further comprises: a third illuminator section operable to illuminate a third section of the reticle radially outwardly of the second section, wherein: the outer dimension of the first section is a second ratio of an inner dimension of the third

section, and the second ratio is 0.05 or less, and, the inner dimension of the second section is a third ratio of the inner dimension of the third section, and the third ratio less than 0.2.
9. A sight device, comprising: an illuminator comprising: a first illuminator section operable to illuminate a first section of a reticle, a second illuminator section operable to illuminate a second section of the reticle radially outwardly of the first section at a spacing from the first section, the second section defining a gap radially outwardly of the spacing, a dimension of the gap is less than an outer dimension of the first section, the dimension of the gap is a ratio of the outer dimension of the first section, and the ratio is less than 0.6.
10. The sight device of claim 9, wherein the ratio is greater than 0.4.
11. The sight device of claim 9, wherein the dimension of the gap is a second ratio of the spacing, and the second ratio is less than 1.3.
12. The sight device of claim 9, wherein the dimension of the gap is a second ratio of an inner dimension of the second section, and the second ratio is less than 0.5.
13. The sight device of claim 9, wherein the illuminator further comprises: a third illuminator section operable to illuminate a third section of the reticle radially outwardly of the second section, wherein the dimension of the gap is a second ratio of an inner dimension of the third section, and the second ratio is less than 0.06.
14. The sight device of claim 9, further comprising a sight housing at least partially supporting the illuminator and a view panel, the sight housing comprising a panel section, and the panel section comprising a plurality of planar outer surfaces oriented to intersect each other and form a plurality of outer edges between adjacent outer surfaces.
15. The sight device of claim 14, further comprising a cover coupled to the sight housing, the cover comprising a cover housing at least partially supporting a second view panel, and the cover housing comprising: a panel section; a first extension extending relative to the panel section; and a second extension extending relative to the panel section, the second extension spaced from the first extension.
16. A non-transitory computer readable medium comprising instructions that when executed cause a plurality of operations to be conducted, the plurality of operations comprising: initiating an illumination of a first section of a reticle; initiating an illumination of a second section of the reticle radially outwardly of the first section after the initiating of the illumination of the first section, the illumination of the second section initiated while the first section is illuminated; ending the illumination of the second section; and initiating an illumination of a point section of the reticle radially outwardly of the second section after the ending of the illumination of the second section, the point section comprising an outer point, and the illumination of the point section initiated while the first section is illuminated.
17. The non-transitory computer readable medium of claim 16, wherein the plurality of operations further comprise: initiating an illumination of a third section of the reticle radially outwardly of the second section after the initiating of the illumination of the point section, the illumination of the third section initiated while the first section and the point section are illuminated, wherein the point section is at least partially aligned with the third section at a radial location.
18. The non-transitory computer readable medium of claim 17, wherein the plurality of operations further comprise: re-initiating the illumination of the second section of the reticle after the initiating of the illumination of the third section, the illumination of the second section re-initiated while the first section, the point section, and the third section are illuminated.
19. The sight device of claim 1, further comprising: a sight housing at least partially supporting the illuminator and a view panel, the sight housing comprising a panel section, and the panel section comprising a plurality of planar outer surfaces oriented to intersect each other and form a plurality of outer edges between adjacent outer surfaces; and a cover coupled to the sight housing, the cover comprising a cover housing at least partially supporting a second view panel, and the cover housing comprising: a panel section; a first extension extending relative to the panel section; and a second

extension extending relative to the panel section, the second extension spaced from the first extension.

20. A sight device, comprising: an illuminator comprising: a first illuminator section operable to illuminate a first section of a reticle, a second illuminator section operable to illuminate a second section of the reticle radially outwardly of the first section at a spacing from the first section, the second section defining a gap radially outwardly of the spacing, a dimension of the gap is less than an outer dimension of the first section; a sight housing at least partially supporting the illuminator and a view panel, the sight housing comprising a panel section, and the panel section comprising a plurality of planar outer surfaces oriented to intersect each other and form a plurality of outer edges between adjacent outer surfaces; and a cover coupled to the sight housing, the cover comprising a cover housing at least partially supporting a second view panel, and the cover housing comprising: a panel section, a first extension extending relative to the panel section, and a second extension extending relative to the panel section, the second extension spaced from the first extension.